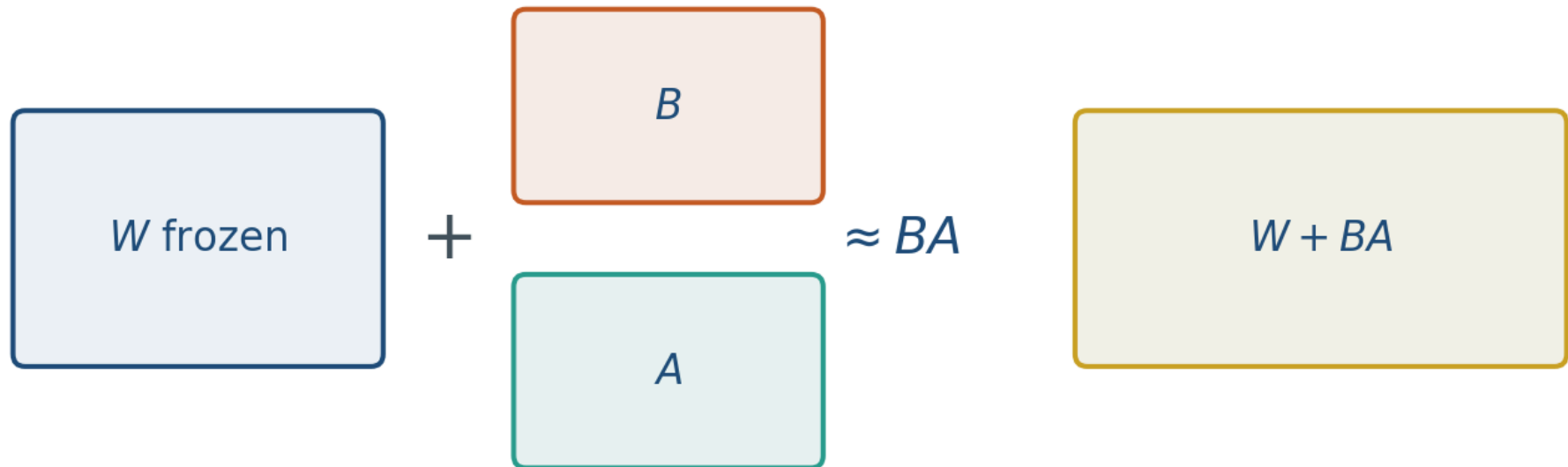


Train (BA) , freeze (W)

- A low-rank pair updates $W + BA$ without rewriting W .
- Rank r is a knob: too small underfits, too large is full FT.

LoRA: train a thin pair BA , leave W alone



Ship adapters, not documents

- Federated sketch: sites train LoRA locally, server averages.
- Still not a privacy proof. It is a smaller delta.

Federated sketch: ship adapters, not raw documents

